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Slim UNETR: Scale Hybrid Transformers to Efficient 3D Medical Image Segmentation Under Limited Computational Resources

Yan Pang, Jiaming Liang, Teng Huang, Hao Chen, Yunhao Li, Dan Li, Lin Huang, Qiong Wang

***Abstract*— Hybrid transformer-based segmentation approaches have shown great promise in medical image analysis. However, they typically require considerable computational power and resources during both training and inference stages, posing a challenge for resource-limited medical applications common in the field. To address this issue, we present an innovative framework called Slim UNETR, designed to achieve a balance between accuracy and efficiency by leveraging the advantages of both convolutional neural networks and transformers. Our method features the Slim UNETR Block as a core component, which effectively enables information exchange through self-attention mechanism decomposition and cost-effective representation aggregation. Additionally, we utilize the throughput metric as an efficiency indicator to provide feedback on model resource consumption. Our experiments demonstrate that Slim UNETR outperforms state-of-the-art models in terms of accuracy, model size, and efficiency when deployed on resource-constrained devices. Remarkably, Slim UNETR achieves 92.44% dice accuracy on BraTS2021 while being 34.4x smaller and 13.4x faster during inference compared to Swin UNETR. Code: https://github.com/aigzhusmart/Slim-UNETR**

***Index Terms*— 3D medical segmentation, lightweight, medical image analysis, resource-limited application**

This work was supported in part by the National Natural Science Foundation of China under Grant 62002074 and Grant 62072452, and the National Key Research and Development Program under Grant 2022YFB4703500, and the Regional Joint Fund of Guangdong (Guangdong–Hong Kong–Macao Research Team Project) under Grant 2021B1515130003, and Regional Joint Fund of Guangdong under Grant 2021B1515120011, and Shenzhen Science and Technology Program under Grant JCYJ20200109115627045, and Shenzhen Science and Technology Innovation Committee Fund under Grant SGDX20210823103201011. (Corresponding authors: Teng Huang and Qiong Wang.)

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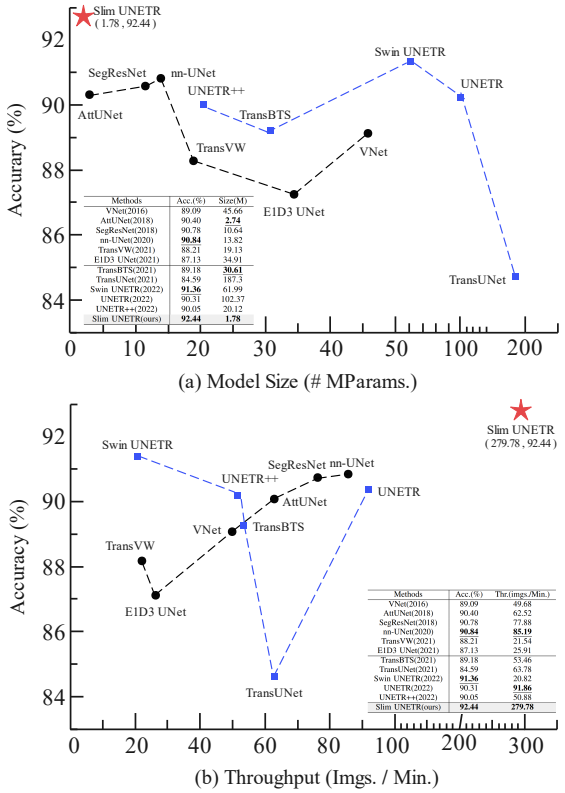


Fig. 1: Slim UNETR demonstrates superior task-level generalization properties compared to other cutting-edge medical segmentation approaches on BraTS2021. The comparison includes a) Model Size vs Accuracy and b) Throughput vs Dice Accuracy. Specifically, Slim UNETR attains a new state-of-the-art Dice accuracy of 92.44% while being 34.4x smaller in size and 13.4x faster in speed than Swin UNETR [1].

I. Introduction

The field of semantic segmentation has experienced a significant increase in interest, driven by the promising technique of emphasizing crucial regions in medical image analysis [2]–[4]. Modern medical segmentation algorithms harness hierarchical representations using advanced multi-scale architectures [5]. In particular, recent hybrid transformer-based methods [6], [7] with distinct U-shaped encoder-decoder

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configurations exhibit scalability for extensive training on high-resolution medical images of various modalities and organs. However, these designs entail considerable complexity and require substantial resources for training and inference. Given the environmental constraints of medical applications and the high-dimensional nature of 3D multimodal medical data, it is essential to prioritize reduced computational costs and high performance in architectural design. Therefore, developing an efficient model tailored for segmentation tasks on resource-constrained medical devices is both vital and necessary.

While current hybrid transformer-based segmentation algorithms [6], [7] have made substantial strides in integrating a transformer into the backbone of the encoder for effective extraction of contextual information, they are faced with the challenge of quadratic complexity due to the self-attention modules. In parallel, CNN-based approaches [8]–[10], while having a more restricted capacity for learning long-range dependencies and smaller regions, provide some relief with their lightweight design, ability to manage fewer parameters effectively, and resilience to data augmentation during the training process. However, they can only extract prominent features in a local context due to their inherent localized receptive fields. This situation poses a question of how to balance the advantages of both transformers and CNNs to develop efficient and high-performance medical segmentation models. In the domain of medical image segmentation, some modern hybrid models [1], [11] are developing ingenious strategies to overcome this challenge. Nevertheless, models that merely integrate transformers or CNNs as encoders or decoders continue to impose substantial computational demands and necessitate extensive data augmentation to forestall overfitting, presenting a substantial challenge. Although these models exhibit an impressive capability for global representation learning and yield remarkable accuracy on segmentation tasks, their deployment in resource-limited settings is still unfeasible. Consequently, there is an urgent call for advancements and inventive solutions that enhance these models' performance and ensure their applicability in resource-restricted circumstances.

This study aims to advance the development of efficient U-shaped medical segmentation frameworks suitable for resource-limited medical units. Such platforms require segmentation models that meet three key qualifications: **high precision, lightweight, and high-inference efficiency**, all of which are critical for resource-limited medical tasks. (i) **High precision** is directly linked to the accuracy of the segmentation model in practical applications. (ii) A **lightweight** model reduces the computational requirements and memory access, making it a more affordable option for modern resource-limited devices with limited resources and smaller storage spaces. In general computer vision, recent segmentation approaches [12]–[14] have already demonstrated that segmentation approaches can effectively learn contextual representations from large-scale datasets. However, to achieve higher precision on high-resolution images and extract stronger feature representations, these models tend to become more complex and larger in size. Some attempts [15]–[17] have been made to reduce the model size by using spatial separable convolutions instead of standard convolutions in hybrid models for 2D natural images. However, these models are still limited in their ability to handle 3D multi-modal medical images like magnetic resonance imaging (MRI) and computed tomography (CT) due to the lack of cross-plane contextual information. While Swin UNETR attempts to address this limitation by using 3D cyclic shifted windows in the self-attention layer, the resulting massive cyclic-shifting operations can lead to high computational costs and low efficiency, which are not suitable for resource-limited devices. (iii) **High-inference efficiency** Achieving high-inference efficiency is crucial for reducing consumption costs in medical resource-limited applications. However, the commonly-used efficiency metric, FLOPs, does not account for the inference efficiency of medical resource-limited devices that have limited factors such as memory access cost and parallelism degree [18]. As a result, FLOPs may not be a suitable metric for evaluating inference efficiency in such devices.

To address the limitations outlined above, we propose an innovative resource-friendly framework, Slim UNETR (**S**lim **UNE**t **TR**ansformers), that can achieve high precision, lightweight, and high-inference efficiency for medical segmentation tasks. The Slim UNETR model combines the strengths of both CNN-based techniques (spatial inductive biases and simple training) and hybrid transformer-based techniques (global procedure) to enhance segmentation accuracy. Our proposed Slim UNETR framework differs from previous research that uses transformers as encoders and standard convolution layers as decoders [6], [19]. Instead, we organize a hierarchical representation using depthwise and transposed 3D convolution operations in both the encoder and decoder. This approach enables the model to gradually capture important local representations for dense prediction, which is illustrated in Figure 2. Our Slim UNETR introduces a novel block designed to cost-effectively aggregate extracted representations within each layer during global processing. The key design of the block is the contextual integration process, which allows for efficient learning of representations through three stages: local aggregation from the surroundings, sparse distribution of authorized tokens, and back diffusion to update representations. With this efficient design, our Slim UNETR achieves better trade-offs between high precision, lightweight, and high-inference efficiency. Figure 1 summarizes the BraTS2021 performance, where our Slim UNETR significantly outperforms current both effective CNN-based and hybrid transformer-based medical segmentation approaches. In particular, our Slim UNETR surpasses the best Swin UNETR dice accuracy of 1.08% but with 34.4x smaller and 13.4x faster on inference. By comparing with the widely used UNETR, our Slim UNETR enhances the segmentation accuracy from 90.31% to 92.44% but with 56.6x smaller and 3.04x faster on inference, as shown in Figure 1. More experiments are conducted to demonstrate the performance of our proposed Slim UNETR by comparing it with more algorithms on three medical datasets, BraTS2021 [20], MSD2019 Task-01 Brain Tumor [21], and MSD2021 Task-08 Hepatic Vessel [22] (See Table I, II).

The main contributions in this paper are summarized as follows:

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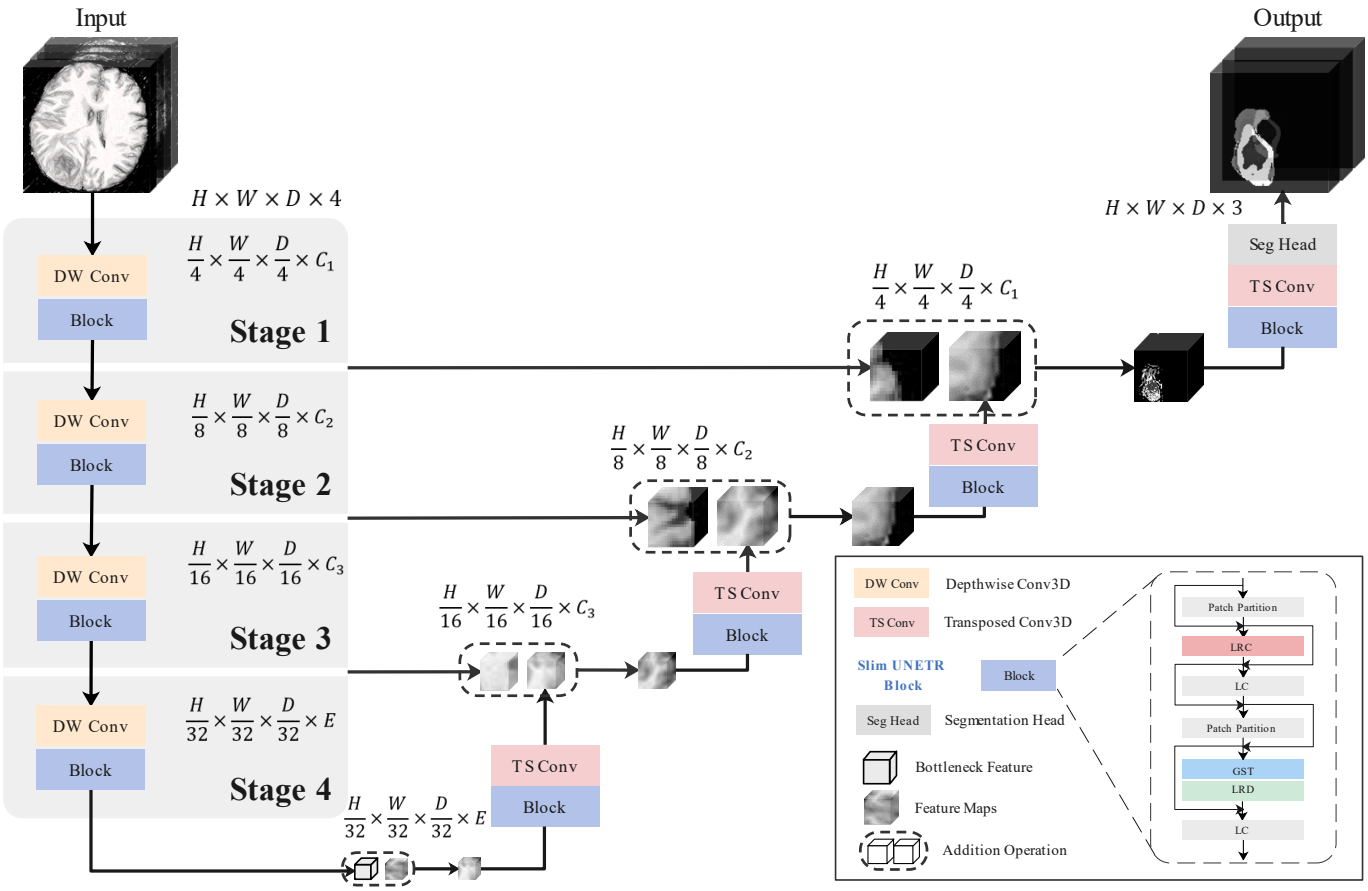


Fig. 2: Overview of the Slim UNETR. Our approach employs a U-shaped encoder-decoder architecture with skip connections to create hierarchical feature maps. Within each stage, a Slim UNETR Block (zoomed in Figure 3) serves as a central component for efficient information exchange, achieved through the decomposition of the self-attention mechanism and cost-effective aggregation of representations.

- A novel resource-friendly framework, Slim UNETR, is proposed for 3D volumetric medical image segmentation, which leverages the merits of both CNN-based and hybrid transformer-based techniques to balance segmentation accuracy and efficiency.
- A Slim UNETR Block with the contextual integration process is designed to efficiently exchange information by decomposing the self-attention mechanism and cost-effectively aggregating extracted representations during global processing.
- Several competing 3D medical segmentation baselines are compared, and it is demonstrated that Slim UNETR achieves new state-of-the-art inference speed and model size while maintaining competitive performance on three public datasets: BraTS2021, MSD2019 Task-01 Brain Tumor, and MSD2021 Task-08 Hepatic Vessel.

II. Related Work

Medical Segmentation Approaches Semantic segmentation plays a vital role in medical image analysis, as it is crucial for examining anatomical structures [23]. Since U-Net [24] first introduces the U-shaped encoder-decoder architecture with convolutional layers, it has shown impressive performance in 3D medical image segmentation tasks [25]–[27]. UNet++ [28] enhances segmentation accuracy by redesigning the skip connections between convolutional blocks to capture information at varying semantic scales. However, CNN-based techniques have limitations in learning long-range dependencies due to the localized receptive fields of convolution operations [6], [29]. UNETR [29] addresses this by incorporating a transformer as the encoder, but this results in quadratic complexity and large model size. Swin UNETR tackles this issue by integrating a Swin Transformer [30] as an encoder and a convolution operation as the decoder via skip connections, reducing model complexity to linear. Nonetheless, Swin UNETR faces challenges in terms of computational load and efficiency due to numerous cyclic-shifting operations. To optimize large-scale hybrid medical segmentation on resource-constrained devices, we propose a factorized self-attention mechanism for cost-effective aggregation of extracted representations.

Lightweight frameworks for resource-limited devices Lightweight frameworks aim to balance accuracy and efficiency for resource-limited hardware. In general computer vision research [31], [32], factorization-based approaches have made CNN-based frameworks more lightweight and resource-friendly. Depthwise separable convolution [33] has been proposed to reduce convolution operation complexity and is

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widely used with residual connections in lightweight CNN-based architectures for resource-constrained vision tasks [34], [35]. With the advent of ViTs [36], which apply transformers to natural image patch sequences using complex multi-head self-attention mechanisms, recent studies [15]–[17] have focused on incorporating lightweight convolutional operations and attention layers to reduce model size. Building on these advances in general computer vision, UNeXt [37] introduces a tokenized convolutional multi-layer perceptron block to decrease the model size and computational complexity for 2D medical resource-limited tasks, while SM-SegNet [38] employs skip connections and pooling indices for 2D tissue segmentation. However, these methods lack cross-plane contextual information when applied to volumetric (3D) medical images, such as CT and MRI. Therefore, it is essential to develop an efficient model tailored for 3D medical resource-limited tasks.

III. Methods

A. Framework Overview

The Slim UNETR framework is a U-shaped architecture, as depicted in Figure 2, comprising an encoder and decoder fused via interconnecting skip connections. The encoder uses convolutional layers to downsample input data with dimensions $H \times W \times D$, and the decoder recovers the data to its original dimensions. Each encoder stage i ($i \in 2, 3, 4$) downsamples features to a size of $\frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times \frac{D}{2^{i+1}}$. The Slim UNETR Block is the key component of this model, incorporating a contextual integration process that facilitates efficient extraction and aggregation of representations from multi-modal feature maps. This is achieved by decomposing the self-attention mechanism and enhancing information exchange. Finally, segmentation probabilities are computed by passing the features from the final step of the decoder through a convolutional layer with an appropriate activation function.

B. Slim UNETR Block

Resource constraints on devices often necessitate the utilization of lightweight models to achieve satisfactory performance in segmentation tasks, as opposed to more substantial alternatives. Consequently, our objective is to develop a lightweight module that can be seamlessly deployed on resource-limited devices. The proposed lightweight Slim UNETR Block, as depicted in Figure 3, encompasses components such as patch partition, local representations congregation (LRC), lineConv, global sparse transformer (GST), and local reverse diffusion (LRD), which are detailed below.

A. Patch Partition

Assuming that a fused 3D multi-modal volume $X : W \times H \times D \times C$ is the input to the Slim UNETR Block, where (W, H, D, C) represent the width, height, depth, and channel of the original resolutions. The patch partitioning layer divides the input volume into N 3D tokens, where $N = HWD/P^3$, and $P = 3$ is the patch size of each token. In this process, a depthwise conv3D operation is used to calculate the embedding of each token, with the kernel and padding size set as 3 and 1, respectively. Unlike standard ViT-based approaches that flatten these fixed-size patches to a sequence of linear embeddings for further self-attention computing at this step, our approach explores local representations first by introducing the image-specific inductive bias in LRC. Moreover, the Patch Partition module, in addition to its role at the input of the Slim UNETR Block, is also applied between the LRC and GST modules to ensure compatibility of feature information with subsequent GST modules.

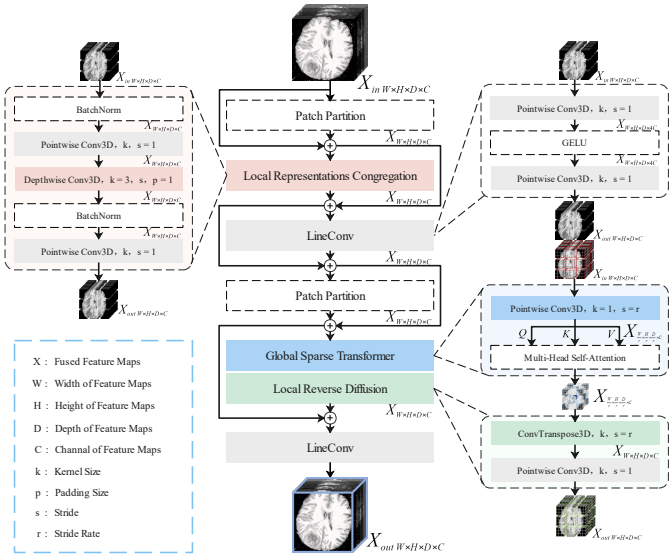


Fig. 3: Structure of Slim UNETR Block. The block illustrates a notable, cost-efficient bottleneck featuring a contextual integration process (zoomed in Figure 4). This term encapsulates the idea of transitioning between local and global representations in order to gain a better understanding of the information being processed.

B. Local Representations Congregation

By examining the internal structures of images, the standard convolutional layer can effectively gather local representations and accurately perform segmentation tasks. However, this process is computationally expensive [15], which is not suitable for resource-limited medical domains. Our proposed LRC leverages the strength of the standard convolutional layer and utilizes fewer parameters to aggregate local representations for resource-limited medical tasks.

As shown in Figure 4 (a), the patches are organized into non-overlapping windows with a size of $W \times W \times W$, where $W = 3$ represents the window size. This process is achieved through the combined point-by-point convolution function of 3D pointwise convolution [39] and the feature aggregation capability of 3D depthwise convolution [40]. Within each local window, information is aggregated from neighboring tokens to the center. To rephrase, the updated center token only integrates signals from locally adjacent tokens, which can be expressed as Equation 1.

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$$
\begin{aligned}
X &= PW(Norm(X_{in}));\\
Y &= DW(X), k = 3, s = 1, p = 1;\\
Z &= PW(Norm(Y));
\end{aligned} \tag{1}
$$

where Norm means batch normalization, which is applied to the standard layer norm, DW and PW denote 3D depthwise convolution and 3D pointwise convolution. The DW utilizes a single 3D convolutional filter per channel, while PW is used to create a linear combination and maintain the same size. The padding size is set to 1, and the filter size is equal to $3 \times 3 \times 3$ in the DW. As a result, the output feature maps retain the same size as the input. With the above work, the LRC model can fully extract local feature information by first convolving point-by-point, then aggregating, and finally convolving point-by-point again. Finally, a residual connection is employed at the end of the LRC module. The whole process of the LRC module is illustrated on the left side of Figure 3.

C. LineConv Module

The LineConv module, which consists of a 2-layer MLP with a GELU activation function in between, is designed to restore the feature map from the aggregated state to the original state so that the feature extraction work of the LRC and GST modules does not affect each other. The output of the LRC module is then fed into a 3D pointwise convolution to increase the number of channels from C to $4C$, followed by another 3D pointwise convolution with GELU activation to reduce the channel size back to C. Moreover, a residual connection is added at the end of the LineConv module. The overall process is shown at the top-right corner of Figure 3.

D. Global Sparse Transformer

The Global Sparse Transformer is employed to structure global representations through windows. This module consists of two designs to sparsify the computational transformer: hierarchical downsampling operations across stages and locally-grouped multi-head self-attention mechanisms (L-MSA) that manage token length and accommodate shared parameters. The former design generates a sparse set of uniformly distributed delegate tokens via a 3D pointwise convolution with a stride rate of r = (4, 2, 2, 1) across different stages. These tokens are subsequently used to compute global sparse attentions among the central tokens in the latter design, as depicted in Figure 4 (b). The global sparse attention coefficients are calculated using Equation 2 [41].

$$
Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{2}
$$

Due to the hierarchically down-sampling operation, the size of the whole tokens is downsampled to $(\frac{W}{r} \times \frac{H}{r} \times \frac{D}{r} \times C)$ across stages. The whole process of GST module is clarified in the middle-right of Figure 3.

E. Local Reverse Diffusion

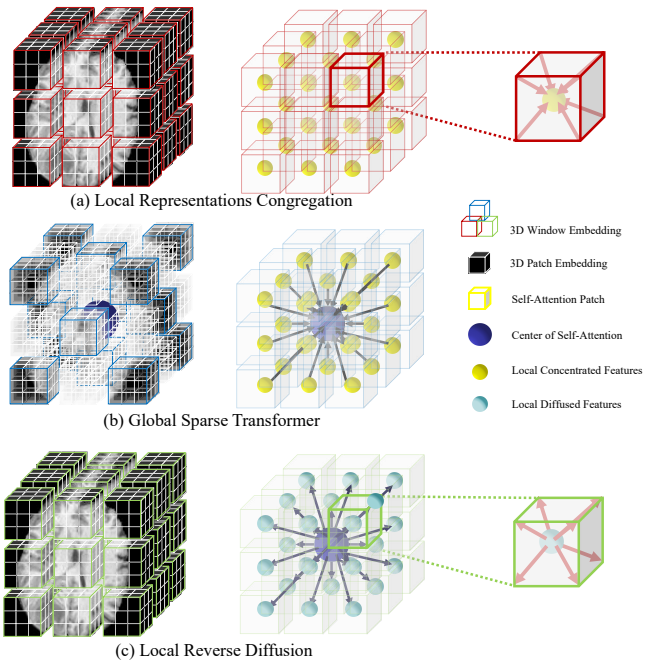


Fig. 4: Contextual integration process. The process aims to improve performance and reduce computational requirements in large-scale segmentation models. It consists of three key components: (1) local representations congregation, which breaks down input data into smaller subsets for easier processing; (2) global sparse transformer, a more efficient self-attention mechanism that learns high-level patterns across local representations; and (3) local reverse diffusion, responsible for integrating global knowledge back into the local representations.

Given that global contextual representations are encoded into center tokens from their neighbors in the GST, it is essential to disseminate this information back to the neighbors. The LRD module serves to distribute the updated knowledge from center tokens to the entire window embedding, as demonstrated in Figure 4 (c). In essence, each local center token can share global representations across windows by employing 3D transposed convolution and pointwise convolution with a consistent hierarchical stride rate of r = (4, 2, 2, 1) across stages. The entire process of the LRD module is depicted at the bottom-right of Figure 3.

C. Contextual Integration Process

The contextual integration process aims to tackle the computational and memory constraints commonly associated with large-scale deep learning models, especially transformers. This method incorporates a resource-efficient bottleneck, concentrating on the effective learning and exchange of representations among its three main components: local representations congregation, global sparse transformer, and local reverse diffusion. By doing so, it enables improved model performance while mitigating the challenges posed by limited resources.

Local Representations Congregation This component is responsible for segmenting input data into smaller, more

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tractable pieces or local representations. By doing so, the model can concentrate on discerning patterns and correlations within these smaller data subsets, thus reducing complexity and computational demands.

Global Sparse Transformer Following the processing of local representations, a global sparse transformer is employed to identify higher-level patterns and connections across various local representations. Sparse transformers utilize a more efficient self-attention mechanism, which decreases memory and computational requirements compared to conventional transformers. This is achieved by attending to a smaller selection of input tokens instead of considering all tokens simultaneously.

Local Reverse Diffusion The final component, local reverse diffusion, is responsible for refining and incorporating the information derived from the global sparse transformer back into the local representations. This step ensures that the local representations are updated with the newly acquired global knowledge, resulting in a more precise and comprehensive understanding of the input data.

As depicted in Figure 4, the contextual integration process contributes to a more economical and efficient learning procedure by decomposing intricate tasks into manageable components, emphasizing local patterns, and employing a sparse transformer to capture global relationships. This approach has the potential to enhance performance, foster better generalization, and decrease computational demands across a range of machine learning and natural language processing tasks.

D. Loss Function

In order to achieve the segmentation task, we adopt soft dice loss [42] as our loss function. Additionally, to address potential issues caused by data imbalance, we utilize focal loss [43] as a fitting technique. These calculations can be expressed using Equation 3 and Equation 4.

$$L_{dice}(G,Y) = 1 - \frac{2}{J}\sum_{j=1}^{J}\frac{\sum_{i=1}^{I} G_{i,j}Y_{i,j}}{\sum_{i=1}^{I} G_{i,j} + \sum_{i=1}^{1} Y_{i,j}} \tag{3}$$

$$\begin{aligned} L_{focal}(G,Y) = -\sum_{j=1}^{J}\sum_{i=1}^{I}(Y_{i,j}G_{i,j}^{2}log(1-G_{i,j}) \\ +(1-Y_{i,j})(1-G_{i,j})^{2}log(G_{i,j})) \end{aligned} \tag{4}$$

where I denotes the number of voxels and J denotes the number of classes; $Y_{i,j}$ and $G_{i,j}$ denote the probability of output and one-hot encoded ground truth for class$j$ at voxel i, respectively.

Formally, we minimize the total loss function by training the total loss function consisting of these double components in the appeal as Equation 5.

$$L_{tot} = \lambda_1 L_{focal} + \lambda_2 L_{dice} \tag{5}$$

where the optimal values of $\lambda_1 = \lambda_2 = 1$ were calculated using a grid-search hyper-parameter optimization.

IV. Experiments and Discussion

In this section, a series of experiments were conducted to evaluate the performance of the proposed Slim UNETR framework on various 3D medical imaging datasets. The computational workload and achieved results were compared with other state-of-the-art approaches in the field to substantiate the excellent trade-off between the efficiency and accuracy of the framework.

A. Datasets

BraTS2021. [20] The BraTS2021 training dataset consists of 1251 data samples, each comprising four 3D MRI modalities: a) native (T1) and b) post-contrast T1-weighted (T1Gd), c) T2-weighted (T2), and d) T2 Fluid-attenuated Inversion Recovery (T2-FLAIR). The data were rigidly aligned and resampled to a 1.0 × 1.0 × 1.0 mm^3 isotropic resolution and skull-deprived. Explanatory notes were provided for three tumor sub-regions: the enhancing tumor, the peritumoral edema, and the necrotic and non-enhancing tumor core. The input image size was set to 240 × 240 × 155. The data were collected from multiple institutions using various MRI scanners, and the explanatory notes were merged into three nested sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

MSD2019 Task-01 Brain Tumor. The No.1 task dataset of Medical Segmentation Decathlon (MSD) 2019 [21] was selected to test the versatility of our approaches, which contains 484 data samples with voxel spacing is 1.0 × 1.0 × 1.0 mm^3, each comprising four 3D MRI modalities that are the same as the BraTS2021 dataset and ground truth labels for glioma segmentation, including necrotic/active tumor and edema, are employed for training the model. Due to the limited training cases, the model performance of representation learning is generally lower than that of models trained with a larger number of labeled cases. In other words, the results of the tumor segmentation task on the brain in MSD2019 can directly reflect the effectiveness of representation learning by comparing them with various approaches.

MSD2021 Task-08 Hepatic Vessel. The No.8 task dataset of MSD2021 [22] was selected to test the versatility of our approaches, which contains 443 CT images of hepatic vessels. Unlike brain data samples, these images present significant differences in terms of anatomical structure and image features. The CT scans were obtained using a variety of parameters, including 120 kVp, exposure time of 500-1100 ms, and tube current of 33-440 mA. The images were reconstructed with a standard convolutional kernel at a section thickness ranging from 2.5 to 5 mm and a reconstruction diameter range of 360-500 mm. The contour annotations were manually adjusted by an expert radiologist specializing in abdominal imaging.

B. Implementation Details

The Slim UNETR was implemented using the MONAI framework and trained on 4 NVIDIA V100 GPUs, while the testing phase utilized an Intel(R) Xeon(R) Gold 6240R CPU. For the BraTS2021 datasets, the four MRI modalities

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TABLE I: Performance comparative analysis of different network architectures for brain tumor segmentation in the BraTS2021 dataset. The performance metrics of Slim UNETR are highlighted in gray, while the best performance in terms of segmentation is recorded in **red**. In addition, we recorded the best values of the most important performance metrics in the CNN-based and hybrid transformer-based methods by underlining. Appr.: Approaches, Metr.: Metrics.

Metr. / Appr.	Dice Similarity Coefficient (%)				Hausdorff Distance 95% (mm)			
	Avg.↑	WT↑	ET↑	TC↑	Avg.↓	WT↓	ET↓	TC↓
TransVW [44]	88.21	92.32	82.09	90.21	8.33	9.89	8.02	7.07
UNet3D [24]	87.93	92.69	84.10	87.10	6.42	8.04	4.90	6.32
E1D3 UNet [45]	87.13	92.42	82.16	86.53	8.18	10.74	7.55	6.24
VNet [42]	89.09	91.38	86.90	89.01	9.83	9.81	10.99	8.69
nn-UNet [46]	90.84	92.71	88.34	91.39	5.33	6.99	4.38	4.62
SegResNet [47]	90.78	92.73	88.31	91.31	5.17	6.33	4.54	4.65
AttUNet [48]	90.40	92.02	88.28	90.94	6.02	8.41	5.23	4.40
Swin UNETR [29]	91.36	93.32	89.08	91.69	**5.03**	**4.64**	4.65	5.81
TransBTS [11]	89.18	91.05	86.75	89.76	6.72	8.54	5.12	6.50
TransUNet [19]	84.59	87.68	83.34	82.75	10.02	12.32	8.04	9.71
UNETR [6]	90.31	92.53	87.59	90.78	6.13	8.97	**4.22**	5.19
UNETR++ [7]	90.05	91.62	86.35	92.17	6.17	6.04	7.46	5.02
VitAutoEnc [49]	76.14	81.41	68.35	78.66	17.92	19.36	16.23	18.19
VIT3D [36]	53.31	53.86	41.16	64.89	29.01	31.51	28.24	27.26
Slim UNETR	**92.44**	**93.77**	**89.36**	**94.19**	5.36	7.34	4.50	**4.24**

for each subject were combined into 4 channels of input, and the corresponding labels were converted into multiple channels based on tumor classes. Data augmentation was applied with probabilities of (0.5, 0.2, and 0.2) for random flips at each axis, intensity scaling, and shifting. For the MSD2021 Task-08 Hepatic Vessel dataset, CT image intensities were clipped to [0, 230] HU to fit the optimal tissue window for hepatic vessel and tumor, and random flip, rotation, and intensity scaling were applied with probabilities of 0.5, 0.25, and 0.5. Depending on the size and characteristics of the image data, 3D patches were randomly cropped out at sizes of 128 × 128 × 128 for two brain datasets and 96 × 96 × 96 for the hepatic vessel dataset, with a positive-to-negative ratio of 1:1. During training, the initial learning rate was set to 0.001, with a linear warmup, cosine annealing learning rate scheduler, and AdamW optimizer with 0.4 weight decay. The Slim UNETR was trained on the BraTS2021 dataset with 1251 and 219 cases in the training and validation sets, on the MSD2019 Task-01 Brain Tumor dataset with 387 and 97 cases in the training and validation sets, and on the Hepatic Vessel dataset with 242 and 61 cases in the training and validation sets, with a training-to-validation sample ratio of 80:20.

C. Evaluation Metrics

For the purpose of lightweight work, we have compiled a set of practical performance metrics that are relevant for running models on resource-constrained devices.

Inference Efficiency is a critical metric that should be optimized and measured in terms of GFLOPs. This metric reflects the model's floating point operations per second and symbolizes the ability to perform a large number of floating point operations in a given time frame.

Model Size is a vital consideration for deployment on resource-limited devices. The model size is typically expressed in terms of MParams and should be affordable for use on resource-limited platforms.

Throughput is an intuitive metric for evaluating the performance of a model in practice. This metric reflects the actual processing efficiency of the model and is an essential metric for assessing the efficacy of lightweight models.

Dice Similarity Coefficient (Dice) quantifies the intersection between the volumetric segmentation predictions and the voxels of the ground truth in Equation 6.

$$Dice = \frac{2 \cdot |\hat{Y} \cap Y|}{|\hat{Y}| + |Y|} \tag{6}$$

where $\hat{Y}$ represents the volumetric segmentation predictions and Y represents the voxels of the ground truth. This coefficient quantifies the intersection between the volumetric segmentation predictions and the voxels of the ground truth, providing a measure of the overlap between the predicted and true regions.

The 95% percent of HD distance distribution (HD95) calculates the 95th percentile of surface distances between the pointsets of ground truth and prediction, which is valuable for evaluating the accuracy and quality of medical image segmentation algorithms in Equation 7.

$$\begin{aligned} HD95 = \max\{&\max_{y' \in Y} \min_{\hat{y}' \in \hat{Y}'} ||\hat{y}' - y'||, \\ &\max_{\hat{y}' \in \hat{Y}'} \min_{y' \in Y} ||y' - \hat{y}'||\} \end{aligned} \tag{7}$$

where $||$ denotes the Euclidean distance function, y and $\hat{y}$ represent individual voxels from the ground truth Y and the predicted segmentation $\hat{Y}$, respectively.

D. Results and Discussion

1) Segmentation Results on Brain Tumor Segmentation Task: In this section, we conducted performance evaluations of the Slim UNETR and other baseline approaches using the BraTS2021 and MSD2019 Task-01 Brain Tumor datasets. This

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TABLE II: Performance comparative analysis of different network architectures for brain and hepatic vessel tumor segmentation in the MSD2019 Task-01 Brain Tumor and MSD2021 Task-08 Hepatic Vessel datasets. The colors of the record in this table are labeled the same as in Table I.

Approaches	MSD2019 Task-01 Brain Tumor (%)				MSD2021 Task-08 Hepatic Vessel (%)		
	Avg↑	WT↑	ET↑	TC↑	Avg↑	Hepatic Vessel↑	Tumors↑
TransVW	73.02	82.12	61.81	75.13	68.62	65.80	71.44
UNet3D	66.43	76.64	56.13	66.52	60.62	57.29	63.95
E1D3 UNet	77.69	88.13	62.39	82.58	66.38	63.55	69.21
VNet	72.95	84.17	61.29	73.39	66.82	63.43	70.21
nn-UNet	77.32	85.83	62.71	83.42	69.12	**66.46**	71.78
SegResNet	78.20	88.68	62.64	83.28	67.52	64.08	71.01
AttUNet	66.46	76.74	54.33	68.33	61.81	56.94	66.68
Swin UNETR	79.16	**90.23**	63.02	84.22	68.95	65.69	72.20
TransBTS	69.61	77.89	57.44	73.51	-	-	-
TransUNet	76.15	86.88	60.36	81.21	65.89	60.95	70.83
UNETR	71.14	78.84	58.49	76.09	66.96	62.84	69.08
UNETR++	79.82	90.07	64.87	84.52	68.51	66.03	70.99
VitAutoEnc	56.20	70.38	36.86	61.36	52.28	48.26	56.30
VIT3D	48.36	62.28	30.06	52.41	48.19	46.32	50.06
Slim UNETR	**80.27**	89.38	**65.70**	**85.73**	**69.73**	65.44	**74.02**

TABLE III: Hardware metrics of brain and hepatic vessel tumor segmentation task in BraTS2021, MSD2019 Task-01 Brain Tumor, and MSD2021 Task-08 Hepatic Vessel datasets, which is recorded in GFLOPs, MParams, and Throughput (images per minute). Since the image sizes of the first two brain-related datasets are the same, they demonstrate comparable values across these three hardware metrics. The colors of the record in this table are labeled the same as in Table I. H.M.: Hardware Metrics.

H.M. / Appr.	Brain Tumor			Hepatic Vessel		
	GFLOPs↓	MParams↓	Throughput↑	GFLOPs↓	MParams↓	Throughput↑
TransVW	2381.3	19.13	21.54	1002.3	19.13	44.4
UNet3D	2382.1	19.24	19.20	1004.7	19.24	40.5
E1D3 UNet	946.23	34.91	25.91	397.13	34.91	49.38
VNet	780.1	45.66	49.68	323.74	45.66	111.54
nn-UNet	446.27	13.82	85.19	185.81	13.82	270.3
SegResNet	330.23	10.64	77.88	137.62	10.64	236.52
AttUNet	166.52	2.74	62.52	69.81	2.74	135.18
Swin UNETR	793.92	61.99	20.82	329.37	61.99	64.26
TransBTS	263.82	30.61	53.46	-	-	-
TransUNet	246.41	187.3	63.78	100.71	177.63	125.52
UNETR	203.21	102.37	91.86	82.41	92.61	131.52
UNETR++	139.44	20.12	50.88	57.51	20.33	75.66
VitAutoEnc	125.66	183.41	193.02	50.13	173.93	223.38
VIT3D	93.78	183.01	118.32	37.54	173.61	192.12
Slim UNETR	**12.42**	**1.79**	**279.78**	**3.81**	**1.79**	**428.16**

allowed us to assess the practicality and effectiveness of our proposed approach for medical image analysis with limited computational resources.

Segmentation Performance We conducted a comprehensive comparison of our proposed Slim UNETR model with other baseline approaches, and the results are presented in Table I. Our proposed model achieved the best performance compared to other baselines, with a top dice accuracy of 92.44% on the BraTS2021 dataset, outperforming the Top 2 model, Swin UNETR, by 1.6%. Moreover, Slim UNETR achieved an impressive average score of 5.36mm for the Hausdorff Distance 95% (HD95), which measures the maximum dissimilarity between the model-segmented region and the ground truth. This score achieved the second highest ranking among the evaluated models, with only a marginal difference of 0.33mm compared to the performance of Swin UNETR. Although the MSD2019 Task-01 Brain Tumor dataset has fewer training data samples available, resulting in a decrease in overall dice accuracy, our Slim UNETR still maintains the Top 1 position with its validation results presented on the left-hand side of Table II. This demonstrates the strong feature extraction capability of our approach, even in scenarios with limited training samples. Benefiting from the contextual integration process, the Slim UNETR allows for efficient information exchange and provides a practical solution for medical imaging analysis on resource-constrained devices.

Model Size CNN-based approaches generally exhibit smaller model sizes in comparison to their hybrid transformer-based counterparts. Among the 13 baselines evaluated, CNN-based frameworks represent 5 of the top 6 models with the smallest sizes, as shown in the left-hand side of Table III. The complex computations introduced by the self-attention

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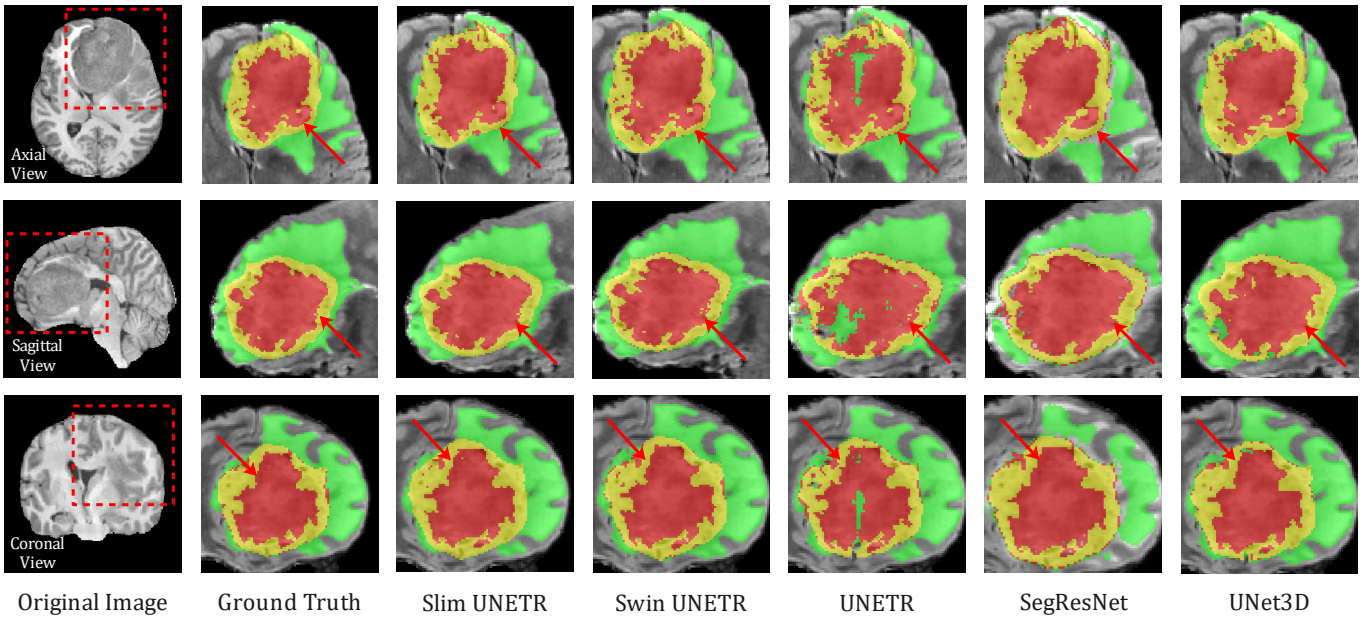


Fig. 5: Qualitative visualizations of the proposed Slim UNETR and baseline approaches. The segmentation of necrotic tumor core (NT, highlighted in green), peritumoral edema (ED, highlighted in yellow), and enhancing tumor (ET, highlighted in red).

mechanism contribute to the larger model sizes commonly associated with hybrid transformer-based ones. For instance, the lightest hybrid model, TransBTS [11], still possesses a parameter count of 30.61 million. However, owing to the design of the Global Sparse Transformer incorporated within the Slim UNETR Block, our proposed approach achieves a model size of a mere 1.79 million, setting a new benchmark in terms of model size reduction.

Inference Efficiency In the introduction, it was highlighted that GFLOPs may not be an ideal metric for evaluating inference efficiency of medical imaging analysis on resource-limited devices as it overlooks important factors like memory access cost and degree of parallelism. To overcome this, throughput has been proposed as a performance metric that quantifies the number of inferences a model can generate in a specific time frame, typically measured in images per minute [50]. In the left-hand side of Table III, although Swin UNETR and VNet show similar FLOPs values, VNet's speed is 2.39 times greater than that of Swin UNETR when assessed by the actual number of images processed per minute. This can be attributed to the differences in the design and operations of the two frameworks. Despite using non-overlapping shifted windows to achieve operational efficiency, its cyclic shift operations pose challenges for deployment, ultimately reducing operational speed compared to VNet. Additionally, while FLOPs values provide some measure of computational operations, they don't necessarily correlate with real-world speed or efficiency. Differences in memory access cost, level of parallelism, and platform characteristics can lead to varied inference speeds, even among models with similar FLOPs, like Swin UNETR and VNet. This discrepancy is particularly critical when considering medical devices with resource limitations. Therefore, we adopt throughput (Thr.) as a more direct metric for inference efficiency, which reflects the model's effectiveness and efficiency in executing tasks on a given hardware configuration.

The left-hand side of Table III illustrates the superior inference efficiency of Slim UNETR compared to the baselines. In the tumor segmentation task on the brain, the throughput of Slim UNETR reaches 279.78, which is 1.45 times higher than one of the second-best approaches, VitAutoEnC. Moreover, Slim UNETR significantly outperforms the VitAutoEnC model in dice accuracy by a margin of 24.07%.

Performance Summary Our Slim UNETR model excels over all baselines in key metrics such as average dice accuracy, model size, and inference efficiency. On the BraTS2021 dataset, it achieves the top-ranked dice accuracy of 92.44%, outperforming Swin UNETR by 1.08%. Furthermore, it surpasses Swin UNETR in throughput by 13.5 times and reduces the model size by 34.4 times. In comparison to the second-best approach, SegResNet, Slim UNETR improves precision by 2.07%, boosts throughput by 3.6 times, and decreases the model size by 5.9 times. Similar results are observed on the tumor segmentation task on the brain in MSD2019. In conclusion, our Slim UNETR model demonstrates a significantly better trade-off between high precision, lightweight, and high-inference efficiency.

2) Segmentation Results on Hepatic Vessel: The right-hand side of Table II presents the tumor and hepatic vessel segmentation results on the MSD2021 Task-08 Hepatic Vessel dataset. Our proposed Slim UNETR model achieves state-of-the-art (SOTA) performance in terms of dice accuracy, model size, and inference efficiency on this dataset. When compared to the second-highest accuracy approach, nn-UNet, Slim UNETR demonstrates considerable advantages with a 0.61% increase in accuracy and a 1.6-fold increase in throughput. Furthermore, in comparison with the second-smallest model size approach, AttUNet, Slim UNETR not only enhances dice accuracy by 7.92% but also decreases GFLOPs by 18.4 times, accompanied by a 3.2-fold increase in throughput. In addition,

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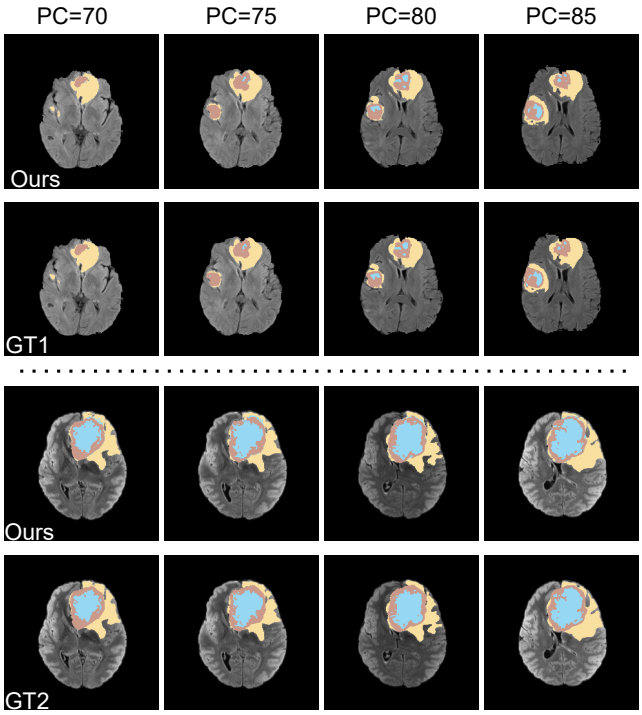


Fig. 6: Qualitative visualizations of the proposed Slim UNETR and ground truth in the BraTS2021 dataset. Compared and evaluated for their respective performances in the segmentation of necrotic tumor core (NT, highlighted in yellow), peritumoral edema (ED, highlighted in red), and enhancing tumor (ET, highlighted in blue).

when compared to Swin UNETR, Slim UNETR achieves an improved trade-off between high precision, lightweight, and high-inference efficiency on the MSD2021 Task-08 dataset. Slim UNETR attains a throughput of 428.16, which is 6.7 times greater than that of the Swin UNETR model, coupled with a 0.78% increase in dice accuracy and a 34.4-fold reduction in model size.

3) Qualitative visualizations of the proposed Slim UNETR and baseline approaches: Figure 5 presents a quantitative visual representation of the segmentation results obtained by different methods for 3D images in the BraTS2021 dataset. This visualization utilizes three standard views - axial, sagittal, and coronal, which are customary in medical imaging. The axial view bisects the body into upper and lower portions, the sagittal view partitions it into left and right sections, and the coronal view segregates it into anterior and posterior sections. By contrasting these three standard views, it becomes apparent that our proposed method manages to accomplish an accurate segmentation of the target. The segmentation closely aligns with the solid representation of the Ground Truth and still maintains an excellent execution speed. We trust these modifications and additions enhance the clarity and comprehensibility of our work.

4) Qualitative visualization with Ground Truth: Figure 6 highlights the performance of the Slim UNETR model in comparison to the ground truth in the BraTS2021 dataset for delineating the primary tumor region, with various colors signifying distinct regions of necrotic tumor core, peritumoral edema, and enhancing tumor. The top two rows exhibit cases of small tumor regions, while the bottom rows present cases of large tumor regions in the brain. The results illustrate that Slim UNETR can achieve distinct boundaries and accurate overlapping areas compared to the ground truth, demonstrating its high-quality 3D segmentation capabilities for medical image analysis. Moreover, we conducted a visual analysis of the segmentation outcomes achieved by the Slim UNETR pair on the MSD2019 Task-01 Brain tumor and MSD2021 Task-08 Hepatic Vessel datasets, while comparing these results with the ground truth and presented them in Figure 7, which can assess the versatility of our proposed model.

TABLE IV: Performance in the BraTS2021 dataset with different settings of Slim UNETR Block. The modules definition: ①: Local Representations Congregation; ②: LineConv; ③: Global Sparse Transformer; ④: Local Reverse Diffusion. We record in **red** and **green** the increase and loss of metrics in ablation. Moreover, the performance metrics of the baseline setting of Slim UNETR are highlighted in gray.

Modules	GFLOPs ↓	MParams↓	Thr.↑	Dice ↑
①	10.71-1.71	0.32 -1.47	346.92 +67.14	83.89 -8.55
①+②	11.23 -1.19	0.71 -1.08	332.98 +53.2	88.67 -3.77
①+(③/④)	11.6 -0.82	0.64 -1.15	320.15 +40.37	90.53 -1.91
①+②+(③/④)	11.72 -0.7	0.91 -0.88	299.81 +20.03	91.16 -1.28
①+②+(③/④)+②	12.42	1.79	279.78	**92.44**

TABLE V: Performance in the BraTS2021 dataset with different settings of specific rates for the different stages in the GST and LRD modules. The colors of the record in this table are labeled the same as in Table IV.

Specific rates	GFLOPs ↓	MParams ↓	Thr. ↑	Dice ↑
(2,2,2,1)	12.34 -0.08	1.73 -0.01	202.5 -77.28	89.84 -2.6
(4,2,2,1)	12.42	1.79	279.78	**92.44**
(4,4,2,1)	12.42 +0.00	1.79 +0.00	299.23 +19.45	85.19 -7.25

E. Ablation Study Results

1) Evaluating the Effectiveness of Slim UNETR Block: We assess the impact of various Slim UNETR Block architectures on medical segmentation performance and make validations on the BraTS2021 dataset. Table IV presents the effectiveness of different layer compositions. With only the Local Representations module, Slim UNETR achieves the lowest computational complexity of 1.07 GFLOPs and the worst dice accuracy of 83.89%, with a throughput of 346.9 images per minute. Adding the GST module (coupled with the LRD module) significantly improves the dice accuracy to 90.53% with a modest throughput reduction. Incorporating the LineCove before and after the GST module further increases accuracy to 92.44% with negligible computational cost. Based on the trade-off between computational resources and dice accuracy, we adopt the Slim UNETR Block composition as described in Section III-B.

2) Determining the Optimal Stride Rates Setting of Stages: We investigate the effect of stride rate settings of different stages in GST and LRD on Slim UNETR performance, and

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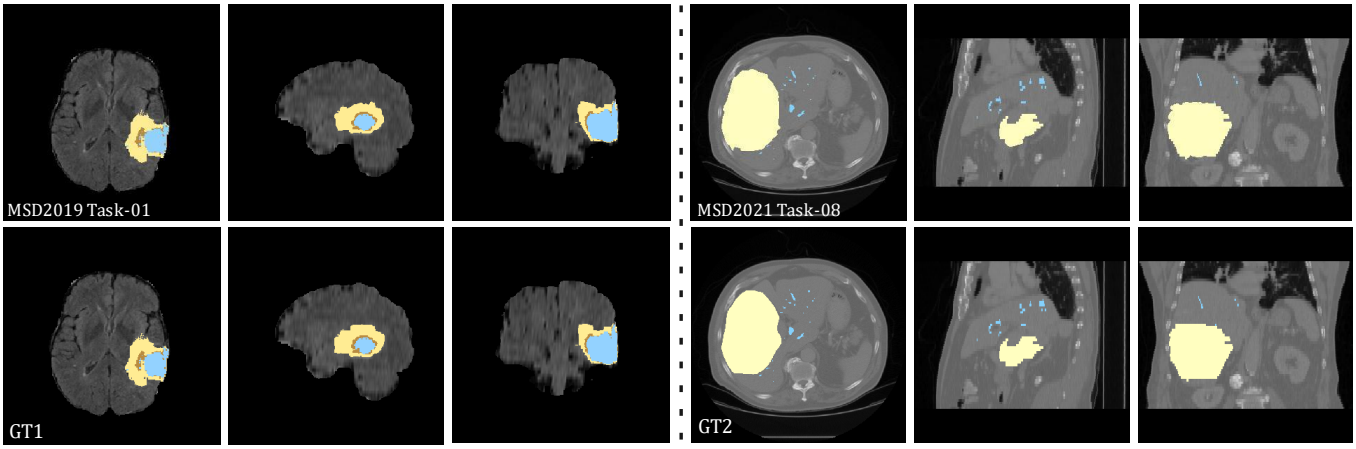


Fig. 7: Qualitative visualizations of the proposed Slim UNETR and ground truth in the MSD2019 Task-01 Brain Tumor and MSD2021 Task-08 Hepatic Vessel datasets. In this table, the highlight colors of tumor segmentation tasks in the brain are the same as Figure 6, while tumor and hepatic vessels are displayed as yellow and blue.

TABLE VI: Performance in the BraTS2021 dataset with different settings of window size in LRC module. The colors of the record in this table are labeled the same as in Table IV.

Window Size	GFLOPs ↓	MParams ↓	Thr. ↑	Dice ↑
1×1×1	12.42 -0.00	1.77 -0.02	231.12 -48.0	88.14 -4.3
3×3×3	12.42	1.79	279.78	**92.44**
5×5×5	12.41 -0.01	1.91 +0.12	280.72 +0.94	87.29 -5.15

TABLE VII: Performance in the BraTS2021 dataset with different numbers of Convs layers and Block layers. The colors of the record in this table are labeled the same as in Table IV.

Num	GFLOPs ↓	MParams ↓	Thr.↑	Dice ↑
2	13.52 +1.1	452.31 +450.52	15.96 -263.82	79.87 -12.57
4	12.42	1.79	279.78	**92.44**
6	16.84 +4.42	80.23 +78.44	34.26 -245.52	88.28 -4.16

make validations on the BraTS2021 dataset, presenting the validation results in Tabel V. It is evident that boosting the stride rate to 4 in stage 2 yields some inference speedup, i.e. 19.45 images per minute increase in throughput, compared to the baseline setting, but can severely affect segmentation accuracy. Conversely, decreasing the stride rate to 2 in stage 1 would result in a significant reduction in inference efficiency, with a throughput loss of 77.28 images per minute. but no demonstrated advantage in segmentation accuracy. Through various considerations, we finally chose to set the stride rate setting of different stages to (4, 2, 2, 1).

3) Determining the Optimal Window Size in LRC module: Employing various window sizes within the LRC module can potentially yield more heterogeneous outcomes. To explore the impact of window sizes on the performance of Slim UNETR, we systematically employ window sizes of 1×1×1, 3×3×3, and 5×5×5 in the LRC module and make validations on the BraTS2021 dataset. Table VI presents the performance impact of different window sizes. From the practical results, although different window sizes do not bring much variation in hardware metrics, applying a 3×3×3 window size is the best choice as it brings the best segmentation performance.

4) Determining the Optimal Number of Stages: The influence of the number of depthwise convolution layers in Slim UNETR on its performance was investigated. The evaluation demonstrated in Tabel VII shows that both reducing the number of stages to 2 and increasing them to 6 adversely affected computational complexity, throughput, and dice accuracy. By setting the number of stages to 4, an optimal balance was achieved, resulting in a throughput of 279.78 images per minute and a dice accuracy of 92.44%, which outperformed other configurations.

V. Limitations and Future Plan

While the Slim UNETR model excels in segmentation tasks and offers a compact model size, its evaluation has been primarily focused on MRI and CT modalities. Further research is needed to examine its applicability and generalizability across a broader range of medical imaging modalities.

VI. Conclusion

In this study, a novel resource-efficient 3D segmentation framework, Slim UNETR, has been introduced, demonstrating the significant potential for resource-constrained medical applications where hardware resources are typically limited. By incorporating the Slim UNETR Block, the efficient exchange of information is facilitated through the contextual integration process, offering a practical solution for medical imaging analysis on devices with limited resources. Additionally, throughput has been employed as an efficiency metric to provide insights into model consumption. Experimental results reveal that Slim UNETR achieves state-of-the-art performance on the BraTS and MSD datasets, suggesting further opportunities for exploration in reducing computational complexity and enhancing inference efficiency.

References

[1] Y. Tang *et al.*, "Self-supervised pre-training of swin transformers for 3d medical image analysis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 20 730–20 740.

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[2] Y. Zhu, J. Cheng, Z.-X. Cui, Q. Zhu, L. Ying, and D. Liang, "Physics-driven deep learning methods for fast quantitative magnetic resonance imaging: Performance improvements through integration with deep neural networks," *IEEE Signal Processing Magazine*, vol. 40, no. 2, pp. 116–128, 2023.

[3] C. Xue *et al.*, "Global guidance network for breast lesion segmentation in ultrasound images," *Medical image analysis*, vol. 70, p. 101989, 2021.

[4] R. Huang *et al.*, "Boundary-rendering network for breast lesion segmentation in ultrasound images," *Medical Image Analysis*, vol. 80, p. 102478, 2022.

[5] L. Li *et al.*, "Automatic multi-plaque tracking and segmentation in ultrasonic videos," *Medical Image Analysis*, vol. 74, p. 102201, 2021.

[6] A. Hatamizadeh *et al.*, "Unetr: Transformers for 3d medical image segmentation," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2022, pp. 574–584.

[7] A. Shaker, M. Maaz, H. Rasheed, S. Khan, M.-H. Yang, and F. S. Khan, "Unetr++: Delving into efficient and accurate 3d medical image segmentation," *arXiv preprint arXiv:2212.04497*, 2022.

[8] B. Koonce, "Mobilenetv3," in *Convolutional Neural Networks with Swift for Tensorflow*. Springer, 2021, pp. 125–144.

[9] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*. PMLR, 2019, pp. 6105–6114.

[10] Y. Pang *et al.*, "Graph decipher: A transparent dual-attention graph neural network to understand the message-passing mechanism for the node classification," *International Journal of Intelligent Systems*, vol. 37, no. 11, pp. 8747–8769, 2022.

[11] W. Wang, C. Chen, M. Ding, H. Yu, S. Zha, and J. Li, "Transbts: Multimodal brain tumor segmentation using transformer," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2021, pp. 109–119.

[12] L. Li, T. Zhou, W. Wang, J. Li, and Y. Yang, "Deep hierarchical semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 1246–1257.

[13] L. Xu, W. Ouyang, M. Bennamoun, F. Boussaid, and D. Xu, "Multi-class token transformer for weakly supervised semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 4310–4319.

[14] A. Ziegler and Y. M. Asano, "Self-supervised learning of object parts for semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 14 502–14 511.

[15] S. Mehta and M. Rastegari, "Mobilevit: light-weight, general-purpose, and mobile-friendly vision transformer," *arXiv preprint arXiv:2110.02178*, 2021.

[16] Y. Pang *et al.*, "Sparse-dyn: Sparse dynamic graph multirepresentation learning via event-based sparse temporal attention network," *International Journal of Intelligent Systems*, vol. 37, no. 11, pp. 8770–8789, 2022.

[17] J. Pan *et al.*, "Edgevits: Competing light-weight cnns on mobile devices with vision transformers," in *European Conference on Computer Vision*. Springer, 2022, pp. 294–311.

[18] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "Shufflenet v2: Practical guidelines for efficient cnn architecture design," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 116–131.

[19] J. Chen *et al.*, "Transunet: Transformers make strong encoders for medical image segmentation," *arXiv preprint arXiv:2102.04306*, 2021.

[20] U. Baid *et al.*, "The rsna-asnr-miccai brats 2021 benchmark on brain tumor segmentation and radiogenomic classification," *arXiv preprint arXiv:2107.02314*, 2021.

[21] A. L. Simpson *et al.*, "A large annotated medical image dataset for the development and evaluation of segmentation algorithms," *arXiv preprint arXiv:1902.09063*, 2019.

[22] M. Antonelli *et al.*, "The medical segmentation decathlon," *Nature communications*, vol. 13, no. 1, p. 4128, 2022.

[23] M. Monteiro *et al.*, "Multiclass semantic segmentation and quantification of traumatic brain injury lesions on head ct using deep learning: an algorithm development and multicentre validation study," *The Lancet Digital Health*, vol. 2, no. 6, pp. e314–e322, 2020.

[24] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.

[25] Q. Dou, H. Chen, Y. Jin, L. Yu, J. Qin, and P.-A. Heng, "3d deeply supervised network for automatic liver segmentation from ct volumes," in *International conference on medical image computing and computer-assisted intervention*. Springer, 2016, pp. 149–157.

[26] L. Yu, X. Yang, H. Chen, J. Qin, and P. A. Heng, "Volumetric convnets with mixed residual connections for automated prostate segmentation from 3d mr images," in *Thirty-first AAAI conference on artificial intelligence*, 2017.

[27] X. Li, H. Chen, X. Qi, Q. Dou, C.-W. Fu, and P.-A. Heng, "H-denseunet: hybrid densely connected unet for liver and tumor segmentation from ct volumes," *IEEE transactions on medical imaging*, vol. 37, no. 12, pp. 2663–2674, 2018.

[28] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "Unet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE transactions on medical imaging*, vol. 39, no. 6, pp. 1856–1867, 2019.

[29] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images," in *International MICCAI Brainlesion Workshop*. Springer, 2022, pp. 272–284.

[30] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 012–10 022.

[31] J. Jin, A. Dundar, and E. Culurciello, "Flattened convolutional neural networks for feedforward acceleration," *arXiv preprint arXiv:1412.5474*, 2014.

[32] S. Mehta, H. Hajishirzi, and M. Rastegari, "Dicenet: Dimension-wise convolutions for efficient networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020.

[33] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1251–1258.

[34] A. Howard *et al.*, "Searching for mobilenetv3," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 1314–1324.

[35] M. Tan *et al.*, "Mnasnet: Platform-aware neural architecture search for mobile," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 2820–2828.

[36] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[37] J. M. J. Valanarasu and V. M. Patel, "Unext: Mlp-based rapid medical image segmentation network," *arXiv preprint arXiv:2203.04967*, 2022.

[38] N. Yamanakkanavar, J. Y. Choi, and B. Lee, "Sm-segnet: A lightweight squeeze m-segnet for tissue segmentation in brain mri scans," *Sensors*, vol. 22, no. 14, p. 5148, 2022.

[39] M. Lin, Q. Chen, and S. Yan, "Network in network," *arXiv preprint arXiv:1312.4400*, 2013.

[40] A. G. Howard *et al.*, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017.

[41] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM computing surveys (CSUR)*, vol. 54, no. 10s, pp. 1–41, 2022.

[42] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *2016 fourth international conference on 3D vision (3DV)*. IEEE, 2016, pp. 565–571.

[43] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2980–2988.

[44] F. Haghighi, M. R. H. Taher, Z. Zhou, M. B. Gotway, and J. Liang, "Transferable visual words: Exploiting the semantics of anatomical patterns for self-supervised learning," *IEEE transactions on medical imaging*, vol. 40, no. 10, pp. 2857–2868, 2021.

[45] S. Talha Bukhari and H. Mohy-ud Din, "E1d3 u-net for brain tumor segmentation: Submission to the rsna-asnr-miccai brats 2021 challenge," *arXiv e-prints*, pp. arXiv–2110, 2021.

[46] F. Isensee, P. F. Jäger, P. M. Full, P. Vollmuth, and K. H. Maier-Hein, "nnu-net for brain tumor segmentation," in *International MICCAI Brainlesion Workshop*. Springer, 2020, pp. 118–132.

[47] A. Myronenko, "3d mri brain tumor segmentation using autoencoder regularization," in *International MICCAI Brainlesion Workshop*. Springer, 2018, pp. 311–320.

[48] O. Oktay *et al.*, "Attention u-net: Learning where to look for the pancreas," *arXiv preprint arXiv:1804.03999*, 2018.

[49] M. J. Cardoso *et al.*, "Monai: An open-source framework for deep learning in healthcare," *arXiv preprint arXiv:2211.02701*, 2022.

[50] D. A. Teich and P. R. Teich, "Plaster: a framework for deep learning performance," *Technology Report TIRIAS Research*, 2018.

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